Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 3.1 · Direct calculations

In an orthonormal basis of the plane, consider the vectors $\vec{u} = (1; -2)$ and $\vec{v} = (4; 3)$.

1. Calculate the scalar product $\vec{u} \cdot \vec{v}$, then the norms $\|\vec{u}\|$ & $\|\vec{v}\|$.
2. Calculate the coordinates of $\vec{u} + \vec{v}$ and deduce $\|\vec{u} + \vec{v}\|^2$.
3. Recover this last result using the identity involving $\vec{u} \cdot \vec{v}$.

Exercise 3.2 · A right-angled triangle?

The plane has an orthonormal frame. We are given the points $A(1; 1)$, $B(4; 2)$ and $C(0; 4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.
2. Deduce that the triangle ABC is right-angled; state at which vertex.
3. Calculate the lengths of the two perpendicular sides, then the area of the triangle in units of area.

Exercise 3.3 · Length without coordinates

The quadrilateral $ABCD$ is a parallelogram such that $AB = 4$, $AD = 3$ and $\widehat{BAD} = 60°$. The lengths are expressed in the same unit.

1. Draw a figure and calculate the scalar product $\overrightarrow{AB} \cdot \overrightarrow{AD}$.
2. Determine the exact lengths of the diagonals $[AC]$ and $[BD]$. Explain how the vectors corresponding to the diagonals are expressed in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
3. Say which diagonal is longer and check that this result is consistent with your figure.

Exercise 3.4 · Find an angle

Two non-zero vectors $\vec{u}$ and $\vec{v}$ satisfy $\|\vec{u}\| = 2$, $\|\vec{v}\| = 6$ and $\vec{u} \cdot \vec{v} = -6$. Let θ be their non-oriented angle, between $0°$ and $180°$.

1. Determine the exact value of $\cos\theta$, then the measure of θ.

2. Explain how the sign of the scalar product predicted that the angle would be obtuse.

Exercise 3.5 · A variable locus

The points A and B are distinct and fixed. Let I be midpoint of $[AB]$. For a real k, we look for the set of points M of the plane that satisfy $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product in terms of MI and AB using Chasles' relation and a scalar identity.
2. Describe precisely the required set according to the value of k. For each case, indicate whether it is empty, a single point or a circle; then give its characteristics.
3. Explain the role of the fact that a squared distance is positive or zero.

Exercise 3.6 · Work done by a force

An object moves in a straight line from A to B over a distance of 4 m. It is subjected to a force constant $\vec{F}$ of magnitude 50 N. The work done by this force is defined by $W = \vec{F} \cdot AB$ and is expressed in joules.

1. Calculate W when the angle between $\vec{F}$ and $\overrightarrow{AB}$ is 60 °.
2. Repeat the calculation for an angle of 120 °.
3. Compare the signs of the two amounts of work and explain, in each situation, whether the force helps the motion or opposes it.

Exercise 3.7 · Contrapositive

We want to prove the following statement, valid for two vectors in the plane:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal} \quad .$$

1. State the contrapositive of this implication.
2. Prove this contrapositive using a scalar identity, then conclude about the original statement.

Exercise 3.8 · Delayed review

A triangle ABC satisfies $AB = 3$, $AC = 4$ and $BC = 5$. We seek $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using any coordinates.

1. Calculate this scalar product by a formula involving the three lengths.
2. Find the result again by determining the nature of the triangle. State the theorem used and its numerical condition.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 250, questions a and b. Calculate the scalar product in the orthonormal b and detail both products of coordinates.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. Use in turn the coordinates, the Chasles' relation and the fact that I is the midpoint.